








Time Series Analysis

Chapter 15: Review and exam preparation



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/3)

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **ASE**: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- **BEKK**: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CCC**: Constant Conditional Correlation (model)
- **DCC**: Dynamic Conditional Correlation
- **DHR**: Dynamic Harmonic Regression
- **DLOG**: Difference of the LOGarithm (EViews function)

Acronyms used in this material (2/3)

- **DM**: Diebold–Mariano (test)
- **EC1**: Error Correction term 1
- **EC2**: Error Correction term 2
- **ECM**: Error Correction Model
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EODHD**: EOD Historical Data (market-data provider)
- **FEVD**: Forecast Error Variance Decomposition
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **GJR**: Glosten–Jagannathan–Runkle (GARCH model)
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HEGY**: Hylleberg–Engle–Granger–Yoo (seasonal unit-root test)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **HP**: Hodrick–Prescott (filter)
- **IPC**: Indicele prețurilor de consum — the consumer price index (CPI)
- **IRF**: Impulse Response Function
- **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- **LB**: Ljung–Box (test)
- **LPPL**: Log-Periodic Power Law

Acronyms used in this material (3/3)

- **LPPLS**: Log-Periodic Power Law Singularity (model)
- **LSTM**: Long Short-Term Memory (network)
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MAPE**: Mean Absolute Percentage Error
- **MASE**: Mean Absolute Scaled Error
- **NBER**: National Bureau of Economic Research
- **OCSB**: Osborn–Chui–Smith–Birchenhall (seasonal unit-root test)
- **OLS**: Ordinary Least Squares
- **PACF**: Partial AutoCorrelation Function
- **PPP**: Purchasing Power Parity
- **QQ**: Quantile–Quantile (plot)
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **SAR**: Seasonal AutoRegressive term (EViews)
- **SARIMA**: Seasonal ARIMA (model)
- **SE**: Standard Error
- **SES**: Simple Exponential Smoothing
- **SIGMASQ**: SIGMA SQuared, the innovation variance (EViews)
- **US**: United States
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax
- **VEC**: Vector (half-vectorised) GARCH model
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: what does the whole course say about a time series, and how is it examined?
  - one chapter that ties Chapters 0–14 together
- **Route**
  - Part I: the course map, two recap slides per chapter, a summary of tests and models
  - Part II: the Box–Jenkins method from start to finish on Romanian inflation
  - Part III: the exam (format, grading, eight solved problems), the team project, attendance and quizzes
- Chapters 11–14 are for self-study



Bucharest University of Economic Studies

Photo: Joe Mabel (2014); CC BY 3.0; Wikimedia Commons

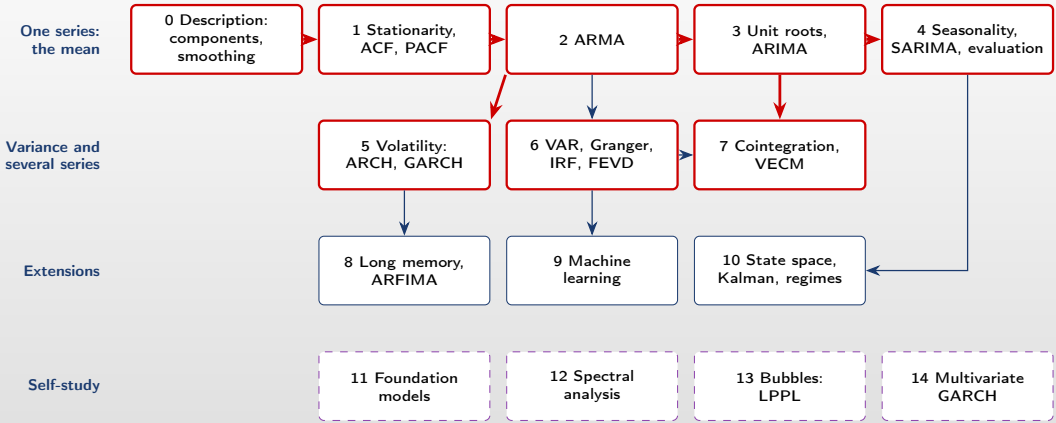
Learning outcomes

- Place every chapter on the path from describing a series to modelling and forecasting it
- Recall the key formula, the key empirical fact and the most common mistake of each chapter
- Choose the right test or model for a given question about a time series
- Run the Box–Jenkins method on a real series: transform, test, identify, estimate, check, forecast
- Read software output (estimates, tests, correlograms) and interpret it in three to five sentences
- Plan the team project, its oral defence and the declaration of AI use

Reading and tools

- Textbook: Huang and Petukhina (2022); free companion: Hyndman and Athanasopoulos (2021)
  - theory: Brockwell and Davis (2016); Hamilton (1994); the method: Box, Jenkins and Reinsel (2008)
- Python Quantlets of this chapter: Quantlets/Ch_15
  - the Box–Jenkins case on the Romanian HICP and the output of exam problem 3
- Lecture notebook, the course in one notebook: open in Google Colab
- Self-assessment: the quiz of this chapter, 20 questions drawn from all chapters

The course map: Chapters 0–14



- Red: the core of the exam, from describing one series to modelling several series together
- Blue: extensions (Chapters 8–10); dashed: self-study (Chapters 11–14), each built on a core chapter: 11 on 9, 12 on 1 and 8, 13 on 3, 14 on 5 and 6

The series of the course

Series	Source	Topic illustrated	Ch.
Romanian real GDP (quarterly)	Eurostat	trend, seasonality, a unit root, regimes	0–4, 10
Romanian HICP inflation (monthly)	Eurostat	persistence, seasonality, long memory	2–4, 8
ROBOR 3M, unemployment	Eurostat	a VAR for the Romanian economy	6
EUR/RON	BNR	a random walk that is hard to beat	0, 3, 5
BET, S&P 500, DAX (daily)	EODHD	volatility clustering, spillovers	1, 5, 6
electricity load (daily, hourly)	ENTSO-E	multiple seasonality, machine learning	4, 9
Nile, sunspots, US yields and GDP	statsmodels, FRED	the classic examples of the textbooks	1, 2, 7, 8, 10

Chapter 0: Introduction: components and exponential smoothing (1/2)

- **Key formulas**
  - additive $y_t = T_t + S_t + R_t$ or multiplicative $y_t = T_t S_t R_t$; the logarithm turns one into the other
  - SES $\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$; MASE = MAE / MAE of the in-sample seasonal naive method
- **Notation**
  - T_t, S_t, R_t: the trend, the seasonal component and the remainder (the irregular part)
  - ℓ_t: the smoothed level at time t; $\alpha \in (0, 1)$: the weight of the newest observation; a large α follows the data closely
  - MAE: the mean absolute error; MASE below 1: better than the seasonal naive method

Chapter 0: Introduction: components and exponential smoothing (2/2)

- **Empirical facts**
  - Romanian GDP grew 2.3 times since 1995 (2.8% a year); Q1 is about -23.8% and Q4 $+20.7\%$ around the trend
  - EUR/RON: $\hat{\alpha}_{SES} = 1.00$, the last value is the best forecast; electricity: Holt–Winters MASE 0.51 against 0.82 for the naive method
- **Common mistakes**
  - judging a forecast on the data used to fit it
  - MAPE for series that cross zero (monthly inflation); an additive model for a seasonal swing that grows with the level

Chapter 1: Stochastic processes and stationarity (1/2)

- **Key formulas**
  - weak stationarity: $EX_t = \mu$, $\mathrm{Cov}(X_t, X_{t+h}) = \gamma(h)$; $\hat{\rho}(h)$ against $\pm 1.96/\sqrt{T}$
  - Ljung–Box $Q^*(m) = T(T+2)\sum_{h=1}^{m} \hat{\rho}(h)^2/(T-h) \sim \chi^2(m)$ Ljung and Box (1978); random walk $\mathrm{Var}(X_t) = t\sigma^2$
- **Notation**
  - μ: the constant mean; $\gamma(h)$: the autocovariance at lag h, which depends only on h; $\hat{\rho}(h)$: the sample autocorrelation; T: the number of observations
  - $Q^*(m)$: the Ljung–Box statistic on the first m lags; a large value (a small p-value) rejects white noise
  - σ^2: the variance of the random-walk shocks; $\mathrm{Var}(X_t) = t\sigma^2$ grows with t, so a random walk is not stationary

Chapter 1: Stochastic processes and stationarity (2/2)

- **Empirical facts**
  - BET returns: $\hat{\rho}_1 = 0.11$, absolute returns $\hat{\rho}_1 = 0.40$ (band ± 0.024): uncorrelated is not independent
  - Romanian GDP: Box–Cox $\hat{\lambda} = 0.02$, the logarithm; one difference too many gives $\hat{\rho}_1 = -0.54$
- **Common mistakes**
  - reading one bar outside the band as a signal (one in twenty is outside by chance)
  - differencing a series that is already stationary

Chapter 2: ARMA models (1/2)

- **Key formulas**
  - $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$; AR(1) $\rho(h) = \phi^h$; MA(1) $\rho(1) = \theta/(1+\theta^2)$
  - AIC $= -2\ln\hat{\mathcal{L}} + 2k$ Akaike (1974), BIC $= -2\ln\hat{\mathcal{L}} + k\ln T$ Schwarz (1978); residual check $Q^*(m) \sim \chi^2(m-p-q)$
- **Notation**
  - L: the lag operator, $LX_t = X_{t-1}$; $\phi(L)$, $\theta(L)$: the AR and MA polynomials of orders p and q; ε_t: white noise
  - $\ln\hat{\mathcal{L}}$: the maximised log-likelihood; k: the number of estimated parameters; a smaller AIC or BIC is better
  - BIC penalises parameters more than AIC, since $\ln T > 2$ for $T \geq 8$

Chapter 2: ARMA models (2/2)

- **Empirical facts**
  - annual growth of Romanian GDP: MA(3) by BIC, a product of overlapping quarters; residual Ljung–Box $p = 0.37$
  - Romanian inflation: AR(2) with $\hat{\phi}_1 + \hat{\phi}_2 = 0.976$, close to a unit root; sunspots: an AR(2) cycle of 10.6 years
- **Common mistakes**
  - reading the AR order from the ACF (it is the PACF that cuts off for an AR)
  - Ljung–Box on residuals with m instead of $m - p - q$ degrees of freedom; a common factor in $\phi(L)$ and $\theta(L)$

Chapter 3: Unit roots and ARIMA models (1/2)

- **Key formulas**
  - ADF: $\Delta y_t = c + bt + \gamma y_{t-\mathbf{1}} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$, $H_{\mathbf{0}}$: $\gamma = 0$ Dickey and Fuller (1979); KPSS: $H_{\mathbf{0}}$ stationarity Kwiatkowski et al. (1992)
  - ARIMA(p, d, q): $\phi(L)(1 - L)^d y_t = c + \theta(L)\varepsilon_t$; forecast variance grows with h when $d \geq 1$
- **Notation**
  - $\Delta y_t = y_t - y_{t-\mathbf{1}}$; c: the constant; bt: the deterministic trend; $\gamma = 0$ means a unit root
  - δ_j: the coefficients of the lagged differences, which remove the autocorrelation of the errors
  - d: the number of differences needed for stationarity; $(1 - L)^d$: the difference of order d; h: the forecast horizon

Chapter 3: Unit roots and ARIMA models (2/2)

- **Empirical facts**
  - log Romanian GDP: $\tau = -2.31$ ($p = 0.43$, 5% critical value -3.45); growth: $\tau = -10.77$; ARIMA(0,1,0) with drift 0.81% a quarter
  - two independent random walks: a significant t in 90% of regressions Granger and Newbold (1974); ADF power only 12.5% for $\phi = 0.95$, $T = 100$
- **Common mistakes**
  - Student or Normal critical values for the ADF statistic
  - reading a non-rejection as proof of a unit root; a regression in levels of $I(1)$ series without a cointegration test

Chapter 4: Seasonality and forecasting (1/2)

- **Key formulas**
  - SARIMA $\phi(L)\Phi(L^s)(1-L)^d(1-L^s)^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$; airline $(0,1,1)(0,1,1)_s$
  - Diebold–Mariano $\bar{d}/\sqrt{\hat{V}/n}$, $d_t = L(e_{1t}) - L(e_{2t})$ Diebold and Mariano (1995), with the HLN correction Harvey, Leybourne and Newbold (1997)
- **Notation**
  - s: the season length (12 for monthly data); $\Phi(L^s)$, $\Theta(L^s)$: the seasonal AR and MA polynomials; D: the number of seasonal differences $(1-L^s)$
  - e_{1t}, e_{2t}: the errors of the two forecasts; $L(\cdot)$: the loss, e.g. the square; d_t: the loss differential; $\bar{d}$: its mean; $\hat{V}$: its estimated long-run variance; n: the number of forecasts
  - under equal accuracy DM is approximately $N(0,1)$: $|\text{DM}| > 1.96$ rejects at 5%

Chapter 4: Seasonality and forecasting (2/2)

- **Empirical facts**
  - airline passengers: MAPE 8.5% for the airline model against 15.5% for the seasonal naive method
  - Orthodox Easter raises Romanian food sales by about 5.9%; daily load: DHR MASE 0.89 against 1.28 (seasonal naive)
- **Common mistakes**
  - comparing AICc of models with different d or D
  - judging forecasts at a single origin; seasonally adjusted data in a SARIMA forecast of raw data

Chapter 5: Conditional volatility: ARCH and GARCH (1/2)

- **Key formulas**
  - GARCH(1,1) $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$ Engle (1982), Bollerslev (1986); $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$, $h_{1/2} = \ln 0.5/\ln(\alpha+\beta)$
  - ARCH-LM $= nR^2 \sim \chi^2(q)$; VaR 1% $= -(\mu + \sigma_{t+1} q_{0.01}(z))$
- **Notation**
  - σ_t^2: the conditional variance; ε_{t-1}: yesterday's shock; $\omega > 0$: the constant; $\alpha \geq 0$: the reaction to shocks; $\beta \geq 0$: the persistence
  - $\bar{\sigma}^2$: the long-run variance, defined only if $\alpha + \beta < 1$; $h_{1/2}$: the half-life of a volatility shock
  - ARCH-LM: R^2 of the regression of ε_t^2 on q of its lags, n observations; $q_{0.01}(z)$: the 1% quantile of the standardised innovations z_t; μ: the conditional mean

Chapter 5: Conditional volatility: ARCH and GARCH (2/2)

- **Empirical facts**
  - GARCH(1,1)-t: $\alpha + \beta = 0.995$ for the S&P 500 (half-life 131 days), 0.991 for the BET (74 days)
  - S&P 500: GJR $\hat{\gamma} = 0.21$ ($t = 10.0$); kurtosis 13.7 for returns, 5.0 for standardised residuals
- **Common mistakes**
  - the half-life written as $1/(1 - \alpha - \beta)$; a long-run variance reported when $\alpha + \beta \geq 1$
  - "VaR 99%" or a negative VaR: the course writes VaR 1%, a positive loss

Chapter 6: VAR models and Granger causality (1/2)

- **Key formulas**
  - VAR(p): $\mathbf{Y}_t = \mathbf{c} + \mathbf{A_1}\mathbf{Y}_{t-\mathbf{1}} + \cdots + \mathbf{A}_p\mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t$ Sims (1980); stable if all eigenvalues of the companion matrix are inside the unit circle
  - Granger $F = [(RSS_R - RSS_U)/p] \,/\, [RSS_U/(T - Kp - 1)]$ Granger (1969); IRF and FEVD with a Cholesky ordering
- **Notation**
  - $\mathbf{Y}_t$: the vector of the K variables; $\mathbf{c}$: the constants; $\mathbf{A}_i$: the $K \times K$ coefficient matrices of lag i; $\boldsymbol{\varepsilon}_t$: the vector of shocks
  - the companion matrix writes the VAR(p) as a VAR(1) of dimension Kp
  - RSS_R, RSS_U: the residual sums of squares without and with the lags of the tested variable; p: the number of restrictions

Chapter 6: VAR models and Granger causality (2/2)

- **Empirical facts**
  - Romanian VAR(2) of growth, inflation and ROBOR 3M: growth Granger-causes ROBOR, $F = 7.26$, $p = 0.001$
  - spillovers among the S&P 500, DAX and BET: 26.4% on average, 57.5% on 17 March 2020
- **Common mistakes**
  - reading Granger causality as causality (it is extra predictability)
  - IRFs reported without the ordering behind them; too many lags for a short sample

Chapter 7: Cointegration and VECM (1/2)

- **Key formulas**
  - Engle–Granger: OLS in levels, then ADF on residuals with MacKinnon critical values Engle and Granger (1987); ECM half-life $\ln 0.5 / \ln(1+\gamma)$
  - VECM $\Delta \mathbf{y}_t = \alpha \beta^\top \mathbf{y}_{t-\mathbf{1}} + \sum_i \Gamma_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t$; trace $-T \sum_{i>r} \ln(1-\hat{\lambda}_i)$ Johansen (1991)
- **Notation**
  - ECM: $\gamma < 0$ is the speed of adjustment towards equilibrium
  - β: the cointegrating vectors (the equilibrium relations); α: the adjustment coefficients; Γ_i: the short-run dynamics; $\mathbf{u}_t$: the shocks
  - $\hat{\lambda}_i$: the estimated eigenvalues, in decreasing order; r: the tested cointegration rank; a trace above its critical value rejects rank r

Chapter 7: Cointegration and VECM (2/2)

- **Empirical facts**
  - US yields at 1, 5 and 10 years: trace $54.6 > 29.8$ for $r = 0$, $3.12 < 3.84$ for $r \leq 2$: rank 2; long-rate ECM half-life 20.8 months
  - PPP for the leu: ADF $p = 0.24$ on the real exchange rate, not stationary; pairs trading on the BVB: Sharpe 0.15 gross, 0.05 net
- **Common mistakes**
  - Dickey–Fuller critical values for Engle–Granger residuals (they must be more negative)
  - a VAR in differences for cointegrated series (it omits the error correction term)

Chapter 8: Long memory and ARFIMA (1/2)

- **Key formulas**
  - $(1-L)^d$ with $\pi_k = \pi_{k-\mathbf{1}}(k-1-d)/k$ Granger and Joyeux (1980), Hosking (1981); $\rho(k) \sim Ck^{\mathbf{2}d-\mathbf{1}}$; $H = d + 1/2$
  - stationary for $d < 1/2$, mean-reverting for $d < 1$; GPH and local Whittle use the lowest m frequencies
- **Notation**
  - d: the order of fractional integration; π_k: the weight of lag k in the expansion of $(1-L)^d$
  - $C > 0$: a constant; $\rho(k) \sim Ck^{\mathbf{2}d-\mathbf{1}}$: the ACF decays hyperbolically, not exponentially
  - H: the Hurst exponent, above 0.5 for long memory, estimated here by the rescaled-range (R/S) statistic; m: the number of low frequencies used by GPH and local Whittle

Chapter 8: Long memory and ARFIMA (2/2)

- **Empirical facts**
  - Romanian monthly inflation: ARFIMA(0,d,0) with $\hat{d} = 0.31$ (SE 0.05) beats ARMA by AIC and BIC
  - S&P 500: H (R/S) 0.53 for returns, 0.86 for $|r_t|$; the Nile: $\hat{d} = 0.40$, but -0.17 after the break of 1898
- **Common mistakes**
  - a break or a regime change read as long memory
  - one bandwidth m only; long memory in returns confused with long memory in volatility

Chapter 9: Machine learning for time series (1/2)

- **Key formulas**
  - direct forecast $\hat{y}_{T+h} = \hat{f}_h(y_T, \ldots, y_{T-p+1}, \mathbf{z}_{T+h})$; ridge $+\lambda \sum \beta_j^2$, lasso $+\lambda \sum |\beta_j|$
  - bias–variance $E(y_0 - \hat{f})^2 = \text{bias}^2 + \text{Var}(\hat{f}) + \sigma^2$; walk-forward validation only
- **Notation**
  - $\hat{f}_h$: the estimated function for horizon h; p: the number of lags used; $\mathbf{z}_{T+h}$: known regressors, e.g. the calendar
  - $\lambda \geq 0$: the strength of the penalty; β_j: the coefficients; ridge shrinks them, lasso sets some exactly to 0
  - y_0: a new observation; σ^2: the noise variance, which no model can remove

Chapter 9: Machine learning for time series (2/2)

- **Empirical facts**
  - S&P 500, the same model: $R^2 = 0.37$ with random k-fold, -0.39 walk-forward: leakage
  - sign of returns: majority class 54.4%, boosting 52.5%; daily load: ridge MASE 0.90, LSTM 1.16 Makridakis, Spiliotis and Assimakopoulos (2020)
- **Common mistakes**
  - random cross-validation and scaling on the whole sample (look-ahead)
  - trees on trending levels (they cannot extrapolate); no simple benchmark in the comparison

Chapter 10: State space models, Kalman filter and Markov switching (1/2)

- **Key formulas**
  - $y_t = Z\alpha_t + \varepsilon_t$, $\alpha_{t+\mathbf{1}} = T\alpha_t + R\eta_t$; $K_t = P_t Z' F_t^{-\mathbf{1}}$, $a_{t|t} = a_t + K_t v_t$ Kalman (1960)
  - Markov switching: durations $1/(1-p_{ii})$, ergodic $\pi_{\mathbf{1}} = (1-p_{\mathbf{22}})/(2-p_{\mathbf{11}}-p_{\mathbf{22}})$ Hamilton (1989)
- **Notation**
  - α_t: the hidden state (e.g. the level); Z, T, R: known system matrices; ε_t, η_t: the measurement and state noise
  - a_t, P_t: the predicted state and its variance; $v_t = y_t - Za_t$: the prediction error, with variance F_t; K_t: the Kalman gain, the weight of the surprise
  - p_{ii}: the probability of staying in regime i; $\pi_{\mathbf{1}}$: the long-run share of regime 1

Chapter 10: State space models, Kalman filter and Markov switching (2/2)

- **Empirical facts**
  - the Nile: $\hat{q} = \hat{\sigma}^2_\eta / \hat{\sigma}^2_\varepsilon = 0.097$, steady-state gain $\bar{K} = 0.267$, the weight of exponential smoothing
  - US GDP: the regimes match the NBER dates in 90% of quarters; Romanian GDP: stable regime 15.2 quarters, volatile 11.2
- **Common mistakes**
  - smoothed probabilities used as if known in real time
  - durations written as $p_{ii}/(1 - p_{ii})$; the last value of the HP gap read as a real-time estimate

Chapter 11: Foundation models for time series (self-study)

- **Key ideas**
  - transformers pre-trained on many series: tokenisation and patching of the values
  - Chronos, TimesFM, Moirai, Lag-Llama: zero-shot forecasts and fine-tuning
  - a fair comparison: the same test sample, MASE, probabilistic scores, simple benchmarks
- **Link with the core of the course**
  - a global model of many series (Chapter 9); judged like any forecast (Chapter 4)
- Chapters 11–14 are for self-study

Chapter 12: Spectral analysis (self-study)

- **Key ideas**
  - the Fourier transform, the periodogram $I(\nu_j)$ and the spectral density
  - smoothed periodograms (Welch), filters (Hodrick–Prescott) and business cycles
  - coherence between two series; wavelets as a time–frequency tool
- **Link with the core of the course**
  - the same information as the ACF, by frequency (Chapter 1); the spectral pole of long memory (Chapter 8)
- Chapters 11–14 are for self-study

Chapter 13: Speculative bubbles: LPPL models (self-study)

- **Key ideas**
  - bubbles as super-exponential growth of prices
  - the LPPLS model: power law plus log-periodic oscillations; estimation and filter conditions
  - the LPPLS confidence indicator on historical bubbles and crashes
- **Link with the core of the course**
  - explosive roots, the opposite of unit roots (Chapter 3); volatility before crashes (Chapter 5)
- Chapters 11–14 are for self-study

Chapter 14: Multivariate GARCH models (self-study)

- **Key ideas**
  - time-varying covariances and correlations; the number of parameters grows fast
  - VEC, BEKK, CCC and DCC; estimation of DCC in two steps
  - dynamic correlations, portfolio variance and VaR 1%
- **Link with the core of the course**
  - GARCH for each series (Chapter 5) and the multivariate view of the VAR (Chapter 6)
- Chapters 11–14 are for self-study

Tests and models (1/2): univariate series and the mean

Question	Tool	Decision	Ch.
Is the series stationary?	plot, ACF; ADF and KPSS	ADF rejects, KPSS does not: stationary	1, 3
Trend: deterministic or stochastic?	ADF and KPSS with a trend	detrend, or difference	3
Is there seasonality?	seasonal plot, ACF at s, $2s$; OCSB, HEGY	Δ_s, or dummies and Fourier terms	4
Which ARMA?	ACF, PACF; AIC, BIC	smallest BIC with white-noise residuals	2
Are the residuals adequate?	Ljung–Box on $m - k$ df; Jarque–Bera	$p > 0.05$: no autocorrelation left	2
Which forecast is better?	test sample, rolling origins; MASE, DM	DM $p < 0.05$; otherwise a tie	0, 4

Tests and models (2/2): variance, multivariate series and extensions

Question	Tool	Decision	Ch.
Does the variance change?	Ljung–Box on r_t^2, ARCH-LM	reject: GARCH(1,1)-t, GJR	5
Does x help to forecast y?	VAR, Granger F test	$p < 0.05$: Granger causality	6
What does a shock do?	IRF, FEVD (Cholesky)	bootstrap bands; try other orderings	6
A long-run equilibrium?	Engle–Granger, Johansen	VECM: β, α, half-life	7
Long memory?	GPH, local Whittle, ARFIMA	several m; check breaks	8
Many predictors, non-linear effects?	ridge, lasso, trees, boosting	walk-forward against a benchmark	9
A hidden level or regime?	Kalman filter, Markov switching	filtered for real time, smoothed for history	10

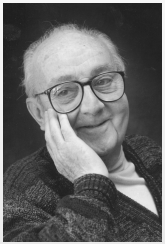
Conventions used throughout the course

- **Transformations**
  - $100 \ln y_t$
  - growth $100\Delta \ln y_t$
  - annual rates $100\Delta_{12} \ln y_t$ (monthly), $100\Delta_4 \ln y_t$ (quarterly)
  - "DLOG(IPC,1,12)" in EViews output is $\Delta\Delta_{12} \ln \text{IPC}$
- **Tests**
  - state H_0, the statistic, its distribution, the critical value or p-value, the decision at 5%
  - then one sentence on what the decision means for the series
- **Model choice**
  - AIC, AICc and BIC only on the same sample and the same d, D
  - forecasts are judged out of sample, against the naive and seasonal naive methods
- **Risk**
  - the level is the tail probability: VaR 1%, a positive loss, $\text{VaR}_\alpha = -q_\alpha$, q_α: the α-quantile of the return

The method in six steps

1. **Plot** and transform: logarithm, Δ, Δ_{12}
2. **Test**: ADF and KPSS for each transformation
3. **Identify**: ACF and PACF of the stationary series
4. **Estimate**: candidate SARIMA models by maximum likelihood; AICc and BIC
5. **Check**: residuals (Ljung–Box, Jarque–Bera, outliers); back to step 3 if they fail
6. **Forecast and evaluate**: a test sample, benchmarks, then intervals

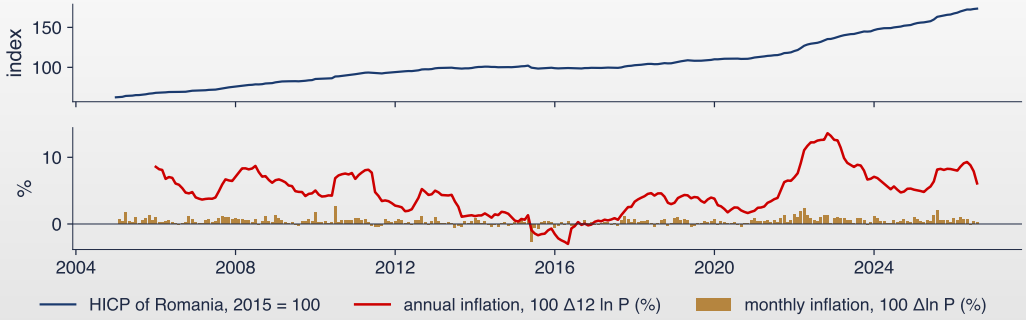
⊡ Data: the HICP of Romania (Eurostat), January 2005 – August 2026, $T = 260$ months; Box, Jenkins and Reinsel (2008)



George E. P. Box (1919–2013)

Photo: DavidMCEddy (2011); CC BY-SA 3.0; Wikimedia Commons

Step 1: the data



⊡ Top: HICP, 2015 = 100; bottom: monthly inflation $100\Delta \ln P_t$ (bars) and annual inflation $100\Delta_{12} \ln P_t$ (line)

TSA_ch15_box_jenkins

Interpreting the data

- **Trend**
  - the index rises almost every month
  - the logarithm is the natural scale
  - monthly inflation averages 0.39%; annual inflation ranges from −3.0% (May 2016) to 13.6% (November 2022)
- **Seasonality**
  - January averages 0.74%, June −0.03% (fresh food)
  - a seasonal pattern in $\Delta \ln P_t$: we will need Δ_{12} or seasonal terms
- **Shocks**
  - tax changes move prices in one month
  - July–August 2025: end of the electricity price caps and a higher VAT; annual inflation from 6.4% to 8.2%

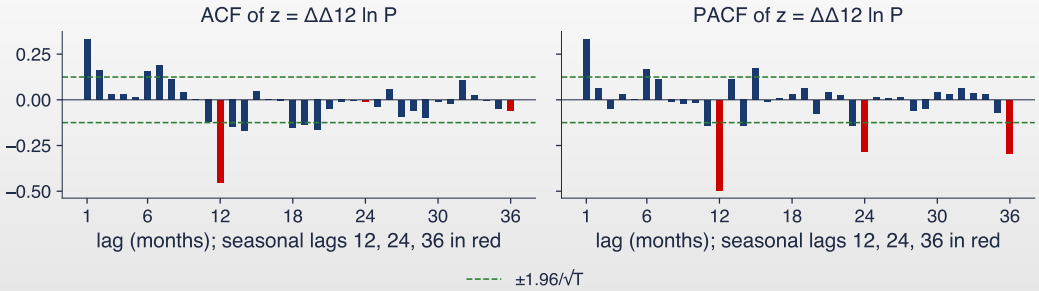
Step 2: unit-root and stationarity tests

Series	**Terms**	ADF τ	p	KPSS η (5%: crit.)	**Verdict**
$100 \ln P_t$	const., trend	-0.96	0.949	0.314 (0.146)	unit root
$\Delta \ln P_t$	const.	-4.00	0.001	0.431 (0.463)	stationary, seasonal
$\Delta_{\mathbf{12}} \ln P_t$	const.	-1.69	0.435	0.424 (0.463)	inconclusive
$z_t = \Delta\Delta_{\mathbf{12}} \ln P_t$	const.	-4.83	< 0.001	0.113 (0.463)	stationary

- ADF: H_0 unit root, lags by AIC Dickey and Fuller (1979); KPSS: H_0 stationarity Kwiatkowski et al. (1992); 5% ADF critical values -3.43 (with trend), -2.87 (constant)
- Annual inflation: neither ADF nor KPSS rejects: an inconclusive result, as in Chapter 3
- Decision: $d = 1$, $D = 1$: we model z_t, both tests agree that it is stationary

TSA_ch15_box_jenkins

Step 3: identification from the correlogram of z_t



- ACF and PACF of $z_t = \Delta\Delta_{12} \ln P_t$, lags 1–36, $T = 247$; seasonal lags in red; band ± 0.125

TSA_ch15_box_jenkins

Interpreting the correlogram

- **Regular part**
  - $\hat{\rho}_1 = 0.33$, $\hat{\rho}_2 = 0.16$
  - PACF $\hat{\phi}_{11} = 0.33$, then small: an AR(1)
  - an MA(1) is also possible: ACF and PACF both shrink after lag 1
- **Seasonal part**
  - ACF -0.45 at lag 12, then -0.01 at 24
  - PACF -0.50, -0.28, -0.29 at 12, 24, 36
  - one seasonal ACF spike and a decaying seasonal PACF: a seasonal MA(1)
- First candidate: SARIMA$(1,1,0)(0,1,1)_{12}$ for $\ln P_t$; the satellites at lags 11 and 13 are the product $\rho_1\rho_{12}$

Step 4: estimating the candidates

Model	param.	AICc	BIC	LB(12) p	LB(24) p
$(1,1,0)(0,1,1)_{\mathbf{12}}$, identified	3	295.2	305.3	0.004	0.004
$(1,1,1)(0,1,1)_{\mathbf{12}}$	4	285.4	298.9	0.029	0.088
$(2,1,1)(0,1,1)_{\mathbf{12}}$	5	282.6	299.4	0.233	0.237
$(1,1,2)(0,1,1)_{\mathbf{12}}$	5	282.8	299.6	0.233	0.217
$(1,1,1)(1,1,1)_{\mathbf{12}}$	5	287.7	304.5	0.017	0.070

- All 36 SARIMA$(p,1,q)(P,1,Q)_{\mathbf{12}}$ with $p,q \leq 2$, $P,Q \leq 1$, on the training sample to August 2024 ($T = 236$); first row: the identified model, then the four other models with the lowest BIC, in order
- param.: all estimated parameters, the variance included; Ljung–Box on the residuals with $m - k$ degrees of freedom, $k =$ the ARMA parameters
- The identified model fails the residual check; the lowest BIC, $(1,1,1)(0,1,1)$, fails at lag 12; the chosen model is $(2,1,1)(0,1,1)_{\mathbf{12}}$, the lowest BIC among the models that pass

TSA_ch15_box_jenkins

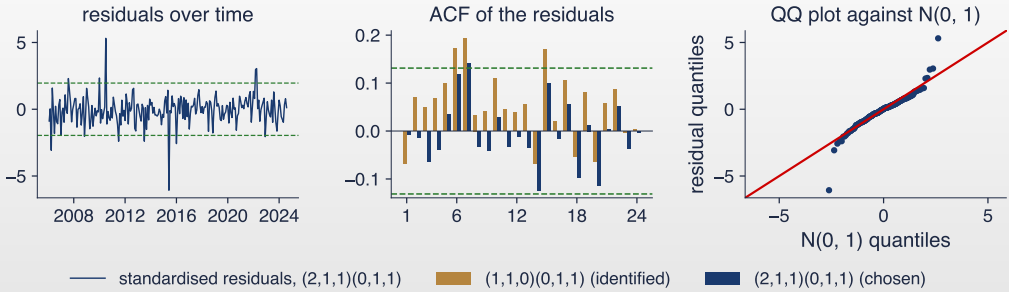
Step 4: the chosen model

Parameter	Estimate	SE	z	p
ϕ_1 (ar.L1)	1.156	0.149	7.76	< 0.001
ϕ_2 (ar.L2)	-0.183	0.122	-1.50	0.133
θ_1 (ma.L1)	-0.856	0.099	-8.61	< 0.001
Θ_1 (ma.S.L12)	-0.935	0.098	-9.55	< 0.001
σ^2	0.179			

- $(1 - \phi_1 L - \phi_2 L^2)\, z_t = (1 + \theta_1 L)(1 + \Theta_1 L^{12})\, \varepsilon_t$, $z_t = \Delta\Delta_{12}\, 100 \ln P_t$
- $\hat{\Theta}_1$ close to -1: the seasonal pattern is almost fixed, Δ_{12} nearly cancels; seasonal dummies would be an alternative
- The smallest AR root has modulus 1.034: inflation shocks are very persistent, as in Chapters 2 and 8

TSA_ch15_box_jenkins

Step 5: checking the residuals



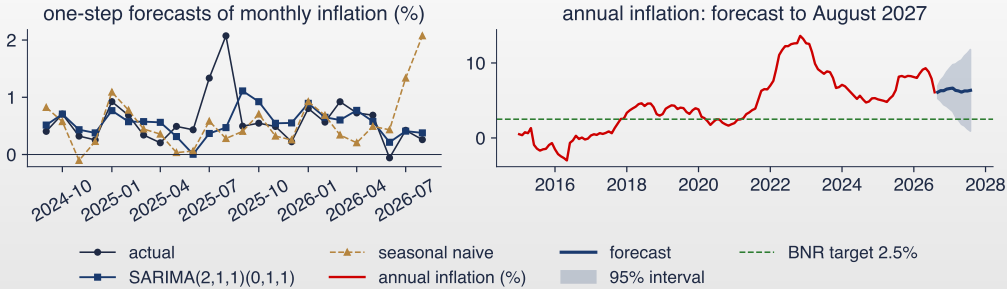
⊡ Standardised residuals of the chosen model; ACF of the residuals of both models; QQ plot against $N(0, 1)$

TSA_ch15_box_jenkins

Interpreting the residual checks

- **No autocorrelation left**
  - Ljung–Box $Q(12) = 10.49$ (8 df, $p = 0.233$), $Q(24) = 24.13$ ($p = 0.237$)
  - the identified model had $p = 0.004$ and 0.004: the extra AR and MA terms reduce the autocorrelation at lags 6–7
- **Not Normal**
  - Jarque–Bera 450, kurtosis 9.9
  - the two largest residuals: June 2015 (-6.1σ, the VAT cut on food) and July 2010 (5.3σ, the VAT increase)
- Consequence: point forecasts are fine, Normal intervals are too narrow when taxes change; dummies for known tax changes would help

Step 6: forecasts



- Left: one-step forecasts of monthly inflation on the test sample (September 2024 – August 2026), parameters fixed at the training estimates
  - right: annual inflation forecasts to August 2027 with 95% intervals, model re-estimated on all data

TSA_ch15_box_jenkins

Interpreting the forecasts

Monthly inflation, test sample	RMSE	MAE	MASE
SARIMA$(2,1,1)(0,1,1)_{12}$	0.45	0.28	0.66
seasonal naive	0.64	0.42	0.99
naive	0.51	0.38	0.88

- SARIMA has the smallest errors, but Diebold–Mariano (HLN) does not reject equal accuracy: -1.40 ($p = 0.174$) against the seasonal naive, -0.47 ($p = 0.642$) against the naive method
  - 24 months are few; the largest misses: August 2025 (2.07% against 0.47%) and July 2025 (1.34% against 0.37%): tax and energy decisions that no univariate model can foresee
- Annual inflation: 6.1% in August 2026; forecast 6.1% for September 2026 [5.3, 6.9], 6.3% for August 2027 [0.9, 11.8]
  - the August 2025 VAT jump leaves the 12-month window in August 2026; the 12-month interval is wide: a near unit root in $\Delta \ln P_t$

TSA_ch15_box_jenkins

Recap: Box–Jenkins on Romanian inflation

- Plot, transform, test: $\ln P_t$ needs $d = 1$ and $D = 1$
- The correlogram suggests $(1,1,0)(0,1,1)_{12}$; the residual check rejects it; the chosen model is $(2,1,1)(0,1,1)_{12}$
- Out of sample the gain over the seasonal naive method is not significant; tax shocks dominate the errors

The written exam: format

- **70% of the final grade**; written, 2 hours
  - five subjects, all compulsory; 9 points for the subjects plus 1 point ex officio
- **Three kinds of subjects**
  - a short derivation: moments, stationarity, the ACF or the form of a process
  - the interpretation of software output: estimates, tests, correlograms, VAR and VECM tables
  - model choice and a forecast computed by hand from the output
- The critical values you need are printed in the output, as in the problems below



The National Institute of Statistics, Bucharest: the source of many exam series

Photo: Dan Mihai Pitea (2009); CC BY-SA 3.0; Wikimedia Commons

Content assessed

- **Chapters 0–10**
  - lectures and seminars
  - the definitions and formulas of the "Prerequisites for Today" slides of every seminar
  - the computations of Part A of the seminars, on paper
- **Reading output**
  - tables like those of Part B and of this chapter
  - EViews or Python output: the same numbers under different names (AR(1), ar.L1; SAR(12), ar.S.L12)
  - no code is written at the exam
- **Core weight**
  - ARIMA and SARIMA, unit roots, VAR and Granger causality, cointegration and VECM, GARCH
  - Chapters 11–14 are for self-study

Grading criteria

- **The method**
  - the formula or the equation, with the numbers substituted
  - a correct method with an arithmetic slip receives most of the points
- **The result**
  - the number with its unit (%, percentage points, months) and its sign
  - for a test: H_0, the statistic, the critical value or the p-value, the decision
- **The interpretation**
  - three to five sentences, statistical and economic
  - a correct number with a wrong interpretation does not receive all the points
  - say what the model cannot show (causality, structural breaks, the future of policy)

Typical problem types

Type	Chapters	Example in this chapter
derivation: moments, stationarity, differencing	1–3	Problem 1: trend plus AR(1) errors
correlogram and unit-root output: identification	1–3	Problem 2: Romanian GDP
SARIMA output and a forecast by hand	2–4	Problem 3: Romanian inflation
VAR and Granger output	6	Problem 4: growth, inflation, ROBOR
cointegration and VECM output	7	Problem 5: US yields
GARCH output and VaR 1%	5	Problem 6: the BET
forecast evaluation; state space and regimes	4, 8–10	Problems 7 and 8

- The eight problems below are review examples with full solutions; the exam has its own subjects
- More practice: Part A of every seminar, Seminar 15 and the practice set (25 problems with numerical answers)

Problem 1: a trend with AR(1) errors

Context

- $Y_t = 10 + 0.5t + u_t$, $u_t = 0.6u_{t-\mathbf{1}} + \varepsilon_t$, $\varepsilon_t \sim WN(0,1)$
- At $T = 100$ we observe $Y_T = 62$

Tasks

1. Compute $E(Y_t)$ and $\mathrm{Var}(Y_t)$ and say whether Y_t is stationary.
2. Show that ΔY_t is an ARMA(1,1) with $\theta = -1$ and compute $\rho_{\Delta Y}(1)$.
3. Forecast $Y_{T+\mathbf{1}}$ and $Y_{T+\mathbf{10}}$ with the correct model and give the variance of the 10-step error.
4. Explain what goes wrong if the analyst differences Y_t.

Problem 1: solution

1. $E(Y_t) = 10 + 0.5t$ depends on t: not stationary; $\text{Var}(Y_t) = 1/(1 - 0.36) = 1.5625$ is constant: **trend-stationary**
2. $\Delta Y_t = 0.5 + \Delta u_t$ and $(1 - 0.6L)\Delta u_t = (1 - L)\varepsilon_t$: ARMA(1,1), $\theta = -1$; $\gamma(0) = 1.25$, $\gamma(1) = -0.25$, $\rho(1) = -0.20$
3. $u_T = 62 - 10 - 50 = 2.0$; $\hat{Y}_{T+\mathbf{1}} = 10 + 50.5 + 0.6 \cdot 2 = 61.70$; $\hat{Y}_{T+\mathbf{10}} = 10 + 55 + 0.6^{\mathbf{10}} \cdot 2 = 65.01$; variance $(1 - 0.6^{\mathbf{20}})/(1 - 0.36) = 1.562$
4. The MA root is 1: ΔY_t is not invertible, $\hat{\rho}_{\mathbf{1}} < 0$ and an MA coefficient near -1 are the signs of over-differencing

Interpretation

- A shock dies out: after 10 periods only 0.012 of the deviation is left, and the error variance stays bounded
- The treatment follows the type of trend: detrend a trend-stationary series, difference a series with a unit root (Chapter 3)

Problem 2: identifying Romanian GDP

Series	Terms	ADF τ	p	crit. 1%; 5%; 10%	KPSS	crit. 5%
$100 \ln Y_t$	c, t	-2.31	0.428	-4.05; -3.45; -3.15	0.198	0.146
$100\Delta \ln Y_t$	c	-10.77	< 0.001	-3.49; -2.89; -2.58	0.263	0.463

- **Context**
  - Romanian real GDP, seasonally adjusted (Eurostat), 2000Q1 – 2026Q2
  - Growth $g_t = 100\Delta \ln Y_t$: ACF -0.05, -0.14; PACF -0.05, -0.14 (lags 1, 2; band ± 0.19); Ljung–Box $Q(8) = 7.28$, $p = 0.506$; $\chi^2_{0.95}(8) = 15.51$
- **Tasks**
  1. Describe the ADF test and decide whether $\ln Y_t$ and g_t are stationary.
  2. Say whether KPSS agrees.
  3. Identify the process with the Box–Jenkins method.
  4. The estimated drift is 0.81 (SE 0.30): write the model and interpret the drift.

Problem 2: solution

1. ADF: $\Delta y_t = c + bt + \gamma y_{t-\mathbf{1}} + \sum \delta_j \Delta y_{t-j} + \varepsilon_t$, $H_\mathbf{0}$: $\gamma = 0$ (unit root); $\ln Y_t$: $-2.31 > -3.45$: not rejected; g_t: $-10.77 < -3.49$: rejected
2. KPSS ($H_\mathbf{0}$: stationarity): $0.198 > 0.146$ rejects for the level, $0.263 < 0.463$ does not for growth: $\ln Y_t \sim I(1)$
3. ACF and PACF of g_t inside the band, $Q(8) = 7.28 < 15.51$: growth is white noise around its mean
4. ARIMA(0,1,0) with drift: $100 \ln Y_t = 100 \ln Y_{t-\mathbf{1}} + 0.81 + \varepsilon_t$, $\hat{\sigma} = 2.47$; $t = 2.72$

Interpretation

- The economy grows by about 0.81% a quarter, about 3.2% a year; quarterly surprises are not predictable
- A shock to the level is permanent: the 2009 and 2020 recessions moved the path of GDP for good

Problem 3: SARIMA output and a forecast by hand

Dependent: z_t	Coefficient	SE	z	p
AR(1)	0.353	0.054	6.53	< 0.001
SAR(12)	-0.488	0.035	-13.90	< 0.001
SIGMASQ	0.254			
Ljung–Box $Q(12)$	22.64			0.012

- **Context**
  - $z_t = \Delta\Delta_{\mathbf{12}}\, 100 \ln \mathrm{HICP}_t$ ("DLOG(IPC,1,12)"), Romania, February 2006 – August 2026, $T = 247$
  - Last values: $z = -1.814$ (August 2026), 0.090 (September 2025), 1.794 (August 2025); annual inflation in August 2026: 6.085%
- **Tasks**
  1. Write the equation of the model and interpret the coefficients.
  2. Test whether the residuals are white noise.
  3. Forecast z and the annual inflation rate for September 2026.

Problem 3: solution

1. $(1-\phi L)(1-\Phi L^{\mathbf{12}})z_t = \varepsilon_t$, so $z_t = \phi z_{t-\mathbf{1}} + \Phi z_{t-\mathbf{12}} - \phi\Phi z_{t-\mathbf{13}} + \varepsilon_t$
2. $\hat{\phi} = 0.353$: a monthly surprise continues one month later; $\hat{\Phi} = -0.488 < 0$: a surprise in a month is partly reversed in the same month a year later; both significant; inverted AR roots 0.35 and 0.94 < 1: stationary
3. Ljung–Box $Q(12) = 22.64$ on $12 - 2 = 10$ df, $p = 0.012$: white noise is rejected at 5%; the model leaves some autocorrelation (compare Step 5 above)
4. $\hat{z}_{T+\mathbf{1}} = 0.353(-1.814) + (-0.488)(0.090) - (-0.173)(1.794) = -0.641 + (-0.044) + 0.310 = -0.375$
5. Annual inflation: $\Delta_{\mathbf{12}} \ln P_{T+\mathbf{1}} = \Delta_{\mathbf{12}} \ln P_T + z_{T+\mathbf{1}} = 6.085 + (-0.375) = 5.71\%$

Interpretation

- Inflation is expected to keep falling in September 2026 as last year's tax shock leaves the 12-month window
- Residual autocorrelation: a richer model (as in Step 4) would change the forecast slightly; a 95% interval needs $\hat{\sigma}$

Problem 4: VAR output and Granger causality

ROBOR equation	coef. [t]		coef. [t]
g_{t-1}	0.142 [3.49]	g_{t-2}	0.074 [1.73]
π_{t-1}	0.240 [2.78]	π_{t-2}	−0.172 [−1.89]
i_{t-1}	1.042 [10.16]	i_{t-2}	−0.119 [−1.20]
constant	−0.109 [−0.55]	R^2	0.93

Null hypothesis	F	p
g does not Granger-cause i	7.26	0.001
π does not Granger-cause i	4.74	0.011
i does not Granger-cause π	0.89	0.417

- **Context**
  - VAR(2) for Romania, 2005Q3 – 2026Q2 ($T = 84$): GDP growth g, annual inflation π, ROBOR 3M i
- **Tasks**
  1. Describe Granger causality and interpret the table.
  2. Write the ROBOR equation and quantify the long-run effect of a permanent 1 pp rise in inflation on ROBOR.
  3. Recompute the first F from $RSS_R = 69.268$ and $RSS_U = 58.273$.

Problem 4: solution

1. x Granger-causes y if the lags of x improve the forecast of y given the lags of y (joint F test) Granger (1969)
   - growth and inflation help to forecast ROBOR ($p = 0.001$ and 0.011); ROBOR does not help to forecast inflation ($p = 0.417$)
2. $i_t = -0.109 + 0.142 g_{t-\mathbf{1}} + 0.074 g_{t-\mathbf{2}} + 0.240 \pi_{t-\mathbf{1}} + (-0.172)\pi_{t-\mathbf{2}} + 1.042 i_{t-\mathbf{1}} + (-0.119) i_{t-\mathbf{2}} + u_t$
3. Long run ($i_t = i_{t-\mathbf{1}}$, $\pi_t = \pi_{t-\mathbf{1}}$): $\partial i/\partial \pi = (0.2402 + (-0.1718))/(1 - 1.0424 - (-0.1186)) = 0.0684/0.0762 = 0.90$ pp
4. $F = \frac{(\mathbf{69.268} - \mathbf{58.273})/\mathbf{2}}{\mathbf{58.273}/\mathbf{77}} = 7.26 > F_{\mathbf{0.95}}(2, 77) = 3.12$: $H_{\mathbf{0}}$ rejected

⊡ **Interpretation**

- The money market rate follows the economy and inflation, as a central bank reaction would imply; the reverse link is not visible in quarterly data
- The long-run effect divides by 0.0762, a number close to zero: it is imprecise; Granger causality is predictability, not proof of a policy effect

Problem 5: cointegration and VECM output

Johansen trace test	$r = 0$	$r \leq 1$	$r \leq 2$
statistic	54.60	21.57	3.12
5% critical value	29.80	15.49	3.84

Adjustment α $[t]$	$\Delta y^{(\mathbf{1})}$	$\Delta y^{(\mathbf{5})}$	$\Delta y^{(\mathbf{10})}$
EC1	-0.021 $[-0.82]$	0.045 $[2.21]$	0.040 $[2.22]$
EC2	0.024 $[0.31]$	-0.122 $[-2.00]$	-0.077 $[-1.43]$

- **Context**
  - US Treasury yields at 1, 5 and 10 years (monthly, FRED), January 1960 – September 2026, VECM with two lagged differences
  - Cointegrating equations: $\text{EC1} = y^{(\mathbf{1})}_{t-\mathbf{1}} - 1.045\, y^{(\mathbf{10})}_{t-\mathbf{1}} + 1.223$; $\text{EC2} = y^{(\mathbf{5})}_{t-\mathbf{1}} - 1.044\, y^{(\mathbf{10})}_{t-\mathbf{1}} + 0.587$
- **Tasks**
  1. Define cointegration and decide the rank.
  2. Write the error-correction part of the equation for $\Delta y^{(\mathbf{10})}_t$ and say which yields adjust.
  3. Interpret the long-run relations economically.

Problem 5: solution

1. $I(1)$ series are cointegrated if a linear combination is $I(0)$ Engle and Granger (1987); trace: $54.60 > 29.80$, $21.57 > 15.49$, $3.12 < 3.84$: rank $r = 2$, one common stochastic trend Johansen (1991)
2. $\Delta y_t^{(\mathbf{10})} = 0.040\,\mathrm{EC1}_{t-\mathbf{1}} + (-0.077)\,\mathrm{EC2}_{t-\mathbf{1}} + \ldots$; on EC1 the 5- and 10-year yields have $|t| > 2$, the 1-year yield $|t| < 1$ on both relations: the short rate does not adjust (weakly exogenous)
3. $\beta \approx 1$: the spreads $y^{(\mathbf{1})} - y^{(\mathbf{10})}$ and $y^{(\mathbf{5})} - y^{(\mathbf{10})}$ are stationary, as the expectations hypothesis implies; half-lives of the equilibrium errors 22.1 and 9.0 months

Interpretation

- The three yields share one trend (the level of rates); the curve can bend, but spreads return to their means
- The long rates do the adjusting: markets move the long end toward the policy-driven short end

Problem 6: GARCH output and VaR 1%

BET	μ	ω	α	β	ν
estimate	0.083	0.0448	0.192	0.799	5.011
SE	0.010	0.0088	0.025	0.024	0.308

- **Context**
  - GARCH(1,1) with Student-t innovations for daily BET log returns in %, since 2000 ($n = 6\,670$)
  - Today: $\sigma_t^2 = 1.0$ and $\varepsilon_t = -3.0$; $t_5^{-1}(0.01) = -3.3649$; about 250 trading days a year
- **Tasks**
  1. Compute the persistence, the half-life and the long-run annual volatility.
  2. Compute σ_{t+1}^2.
  3. Compute the one-day VaR 1% with the standardised t_5 quantile and compare it with the Normal one.

Problem 6: solution

1. $\alpha + \beta = 0.9907$; $h_{\mathbf{1/2}} = \ln 0.5 / \ln 0.9907 = 74$ days; $\bar{\sigma}^{\mathbf{2}} = 0.0448/(1 - 0.9907) = 4.82$, $\sqrt{250 \times 4.82} = 34.7\%$ a year
2. $\sigma^{\mathbf{2}}_{t+\mathbf{1}} = 0.0448 + 0.19162 \times 9 + 0.7991 \times 1.0 = 0.0448 + 1.7246 + 0.7991 = 2.5685$, $\sigma_{t+\mathbf{1}} = 1.603\%$
3. $q_{\mathbf{0.01}}(z) = -3.3649 \times \sqrt{3/5} = -3.3649 \times 0.7746 = -2.606$; VaR 1% $= -(0.083 + 1.603 \times (-2.606)) = 4.09\%$; Normal: 3.65%

Interpretation

- Persistence close to 1: a shock to volatility halves only after about 74 trading days
- The long-run volatility (34.7%) is far above the sample one (22.1%): when $\alpha + \beta \approx 1$, $\omega/(1 - \alpha - \beta)$ is very imprecise
- The factor $\sqrt{3/5} = \sqrt{(\nu - 2)/\nu}$ rescales the $t_{\mathbf{5}}$ quantile to innovations of variance 1
- Heavy tails raise VaR 1% above the Normal value: a loss larger than 4.09% is expected on one day in 100

Problem 7: evaluating forecasts

Test month	1	2	3	4	5	6
errors, SARIMA	0.3	−0.5	0.2	0.6	−0.4	0.1
errors, seasonal naive	0.5	−0.7	0.6	0.4	−0.9	0.3

Context

- Monthly inflation forecasts on a test sample of six months; the in-sample MAE of the seasonal naive method is 0.6
- A colleague reports $R^2 = 0.37$ for a boosting model of daily S&P 500 returns, with random 5-fold cross-validation

Tasks

1. Compute RMSE, MAE and MASE for both methods.
2. Compute the Diebold–Mariano statistic on squared errors, $\bar{d}/(s_d/\sqrt{n})$, and decide at 5% ($t_{0.975}(5) = 2.571$).
3. Comment on the colleague's R^2.

Problem 7: solution

1. SARIMA: RMSE 0.389, MAE 0.350, MASE 0.58; seasonal naive: RMSE 0.600, MAE 0.567, MASE 0.94
2. $d_t = e_{A,t}^2 - e_{B,t}^2$ (A: SARIMA, B: naive): -0.16, -0.24, -0.32, 0.20, -0.65, -0.08; $\bar{d} = -0.208$, $s_d = 0.281$; DM $= -1.82$, $|-1.82| < 2.571$: equal accuracy is not rejected
3. Random folds put future days in the training set: leakage; walk-forward validation gives $R^2 = -0.39$ for the same model (Chapter 9)

Interpretation

- MASE below 1: SARIMA beats the seasonal naive method on average, but six months cannot show that the gain is real
- Only out-of-sample, time-ordered evaluation counts as evidence of forecasting skill

Problem 8: state space, regimes and long memory

- **Context**
  - Local level model: $\sigma_\varepsilon^2 = 4$, $\sigma_\eta^2 = 1$; prediction $a_t = 10$, $P_t = 2$; observation $y_t = 13$
  - Romanian GDP growth, two regimes: $p_{11} = 0.9108$ (volatile), $p_{22} = 0.9341$ (stable) (Chapter 10)
  - Romanian monthly inflation: ARFIMA(0,d,0) with $\hat{d} = 0.31$ (SE 0.05) (Chapter 8)
- **Tasks**
  1. Run one Kalman update and prediction; compute the steady-state gain and the equivalent SES weight.
  2. Compute the expected durations of the two regimes and the long-run share of the volatile regime.
  3. Compute H and $\rho(1)$ for the ARFIMA model and say whether inflation is stationary and mean-reverting.

Problem 8: solution

1. $F_t = 2+4 = 6$, $K_t = 2/6 = 0.333$; $a_{t|t} = 10 + 0.333 \times 3 = 11.00$, $P_{t|t} = 2(1-0.333) = 1.333$; $a_{t+\mathbf{1}} = 11.00$, $P_{t+\mathbf{1}} = 2.333$
2. $q = \sigma^{\mathbf{2}}_{\eta}/\sigma^{\mathbf{2}}_{\varepsilon} = 0.25$: $\bar{P}/\sigma^{\mathbf{2}}_{\varepsilon} = (q + \sqrt{\mathbf{q^2 + 4q}})/2 = 0.6404$, $\bar{K} = 0.6404/(1+0.6404) = 0.390 = \alpha_{SES}$
3. $E(D_{\mathbf{1}}) = 1/(1-0.9108) = 11.2$ quarters, $E(D_{\mathbf{2}}) = 15.2$ quarters; $\pi_{\mathbf{1}} = (1-0.9341)/(2-0.9108-0.9341) = 0.425$
4. $H = d + 0.5 = 0.81$; $\rho(1) = d/(1-d) = 0.45$; $0 < d < 0.5$: stationary with long memory, mean-reverting

Interpretation

- The filter moves the level one third of the way to the surprise; in the long run the weight is that of exponential smoothing
- In the long run Romanian growth is in the volatile regime about 42% of the time: transition, 2009 and 2020
- Inflation shocks fade slowly, hyperbolically: forecasts return to the mean more slowly than those of an AR model

The team project: content and deliverables

- **20% of the final grade**
  - a team of 2–4 students analyses real series with the methods of the course
  - one concrete question: forecasting EUR/RON, modelling Romanian inflation, monetary policy and bank rates
  - one series: trend and stationarity, smoothing, ARIMA or SARIMA, a training set, a test set and a horizon, point and interval forecasts, two methods compared
  - several series: unit roots, cointegration, VAR or VECM, Granger causality, IRF and FEVD
- **Deliverables**
  - a repository whose code reproduces every number and every chart from the data
  - a short report: the question and the literature, data sources, models, conclusions, references
  - a presentation; the file `AI_USE.md`

Project grading criteria and the oral defence

- **The question**
  - clear, answerable with the data, useful to someone
  - "Can a SARIMA forecast Romanian inflation better than the seasonal naive method?" is a question; "an analysis of inflation" is not
- **Methods, checks, interpretation, reproducibility**
  - the right test for the question (the summary of tests and models); residual diagnostics; out-of-sample evaluation against a benchmark
  - what the numbers mean and what the data cannot show; the repository runs from the data to every chart
- **The oral defence**
  - the project is graded only after its presentation
  - each member explains the code and the results: what does this line compute? why this test and not another?
  - a line of code that cannot be explained does not count as your own work, with or without AI

The file AI_USE.md

- **AI tools are allowed and must be declared**
  - for every use: the tool, the prompt, what was kept and what was corrected
  - the errors of the AI tool that you found and how you found them
- **You are responsible for every number and every reference**
  - every number: recomputed with your own code from the data
  - every reference: its DOI opened and its title checked
- The typical AI errors of this course: the slide “Necessary checks” of the section on AI

The final grade

Component	Weight	When	Criteria
written exam	70%	exam session	method, result, interpretation
team project	20%	end of semester	question, methods, checks, oral defence
attendance and quizzes	10%	every week	presence at the lectures and seminars; the chapter quizzes

- The seminars are for practice and are not graded; students hand in nothing from the seminars
- The quizzes on the course site are taken with the ASE account; name and score are recorded for the grade

Possible contribution of AI

- **Practice**
  - new exercises in the style of the exam, with output generated from real data by your own code
- **Explanation**
  - a second explanation of an output table, a derivation or a test you did not understand
- **Project**
  - a first draft of the code of the Box–Jenkins steps, of a rolling evaluation, of a VECM
- Example prompt
  - `Write Python code that downloads the Romanian HICP from Eurostat (prc_hicp_minr, M.I15.TOTAL.RO), fits every SARIMA(p,1,q)(P,1,Q)12 with p, q <= 2 and P, Q <= 1 to 100 log HICP up to August 2024, reports AICc, BIC and Ljung-Box p-values with m - k degrees of freedom, and compares one-step forecasts with the seasonal naive method by MASE and the Diebold-Mariano test.`

Necessary checks

- The null hypotheses: ADF has a unit root under H_0, KPSS has stationarity; an answer that swaps them is wrong
- The degrees of freedom: Ljung–Box on residuals uses $m - k$; Granger F uses $(p, T - Kp - 1)$
- The signs: the EViews and Python conventions for MA terms and for the error-correction coefficient
- The conventions: AIC and BIC only for the same d and D; VaR 1%, never "VaR 99%"
- The time order: no information from the test sample in the estimation, the scaling or the model choice
- Every number recomputed with your own code; every reference checked by its DOI

Key takeaways

- Look at the data first: components, transformations, the correlogram
- Decide d and D with ADF and KPSS together; identify with ACF and PACF; choose with BIC; check the residuals
- Model the variance when it clusters (GARCH); model several series together with VAR or VECM
- Judge forecasts out of sample against simple benchmarks; a better in-sample fit is not a better forecast
- The exam rewards method, result and interpretation; the project rewards a clear question answered honestly

Self-assessment

- **Question**: ADF does not reject and KPSS rejects. What do you conclude?
  - **Answer**: both point to a unit root: difference the series
- **Question**: the residuals of an ARMA(1,1) give Ljung–Box $Q(10) = 16$. Is the model adequate at 5%?
  - **Answer**: with $10 - 2 = 8$ df the critical value is 15.51: no, autocorrelation is left
- **Question**: two $I(1)$ series are cointegrated. Why not a VAR in differences?
  - **Answer**: it omits the error-correction term and loses the long-run information

References (1/3)

- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723. doi:10.1109/TAC.1974.1100705
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (2008). *Time Series Analysis: Forecasting and Control* (4th ed.). Wiley. doi:10.1002/9781118619193
- Brockwell, P. J., & Davis, R. A. (2016). *Introduction to Time Series and Forecasting* (3rd ed.). Springer. doi:10.1007/978-3-319-29854-2
- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431. doi:10.1080/01621459.1979.10482531
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. doi:10.2307/1912773
- Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276. doi:10.2307/1913236

References (2/3)

- Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3), 424–438. doi:10.2307/1912791
- Granger, C. W. J., & Joyeux, R. (1980). An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1), 15–29. doi:10.1111/j.1467-9892.1980.tb00297.x
- Granger, C. W. J., & Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, 2(2), 111–120. doi:10.1016/0304-4076(74)90034-7
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384. doi:10.2307/1912559
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291. doi:10.1016/S0169-2070(96)00719-4
- Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176. doi:10.1093/biomet/68.1.165
- Huang, C., & Petukhina, A. (2022). *Applied Time Series Analysis and Forecasting with Python*. Springer. doi:10.1007/978-3-031-13584-2

References (3/3)

- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts. otexts.com/fpp3
- Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580. doi:10.2307/2938278
- Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1), 35–45. doi:10.1115/1.3662552
- Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), 159–178. doi:10.1016/0304-4076(92)90104-Y
- Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303. doi:10.1093/biomet/65.2.297
- Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020). The M4 Competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1), 54–74. doi:10.1016/j.ijforecast.2019.04.014
- Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2), 461–464. doi:10.1214/aos/1176344136
- Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1–48. doi:10.2307/1912017